5. **Answer: C**

In an inelastic collision, the ball loses some of its kinetic energy and its momentum upon rebound is lower in magnitude than its momentum just before contact with the surface.

Option A: Not possible. Change in momentum of $0.5p$ suggests the ball continues in the same vertically downward direction even after hitting the surface.
Option B: Not possible. Change in momentum of $1.0p$ suggests the ball comes to rest after hitting the surface.
Option C: Possible. Change in momentum of $1.5p$ suggests the magnitude of ball's momentum after rebound is $0.5p$. Hence, change in momentum (taking upward as positive) is $0.5p - (-1.0p) = 1.5p$.
Option D: Not possible. Change in momentum of $2.0p$ suggests the ball collides elastically with the surface with no loss in kinetic energy.